CS383 · SOFTWARE ENGINEERING

Midterm 462 Solutions

20 marks

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01 Solution Q1: MCQ

#	Answer	Reason	Lecture
1	**a**	Development process.	5
2	**b**	Understandability.	5
3	**b**	Continuous feedback. XP is the opposite of long-term planning and heavy documentation.	4
4	**a**	Refactoring: continuously improving the code without changing what it does.	4

02 Solution Q2: ethics

SEEPP ethic example	Principle	Keyword
Give a fair hearing to the opinions, concerns, or complaints of a colleague.	**COLLEAGUES**	colleague
Improve their knowledge of this Code, its interpretation, and its application to their work.	**SELF**	improve their knowledge
Ensure that software engineers are informed of standards before being held to them.	**MANAGEMENT**	informed of standards
Ensure an appropriate method is used for any project on which they work or propose to work.	**PRODUCT**	appropriate method
Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.	**JUDGMENT**	conflicts of interest
Maintain the integrity of data, being sensitive to outdated or flawed occurrences.	**PRODUCT**	integrity of data
Accept full responsibility for their own work.	**PUBLIC**	responsibility for their own work
Not give unfair treatment to anyone because of any irrelevant prejudices.	**SELF**	prejudices

Watch these two

The conflicts of interest statement is **JUDGMENT**, not Management. The integrity of data statement is **PRODUCT**, not Profession.

03 Solution Q3: FR and NFR

Functional
1. The system will allow employees to create tasks with a title, description, priority level, and due date.
2. Employees can update the status of their tasks.

Non-functional	Kind
1. The system should handle up to 100 concurrent users without delays.	Performance
2. Ensure that user data is encrypted during transmission.	Security

A third valid FR is: assign tasks to team members.

04 Solution Q4: the four activities

1) Software specification
2) Software development
3) Software verification and validation
4) Software evolution

05 Solution Q5: development model

Project	Answer	Justification
1	**Plan-Driven**	The requirements are stable and known from the start, the stakeholders are not available for feedback, and the teams are large, distributed and communicate through documentation. So all the specification and design are fixed before implementation.
2	**Agile**	The stakeholders are continuously involved, the requirements change often, and the team is small (5) and co-located with daily stand-ups. Agile welcomes change and relies on early feedback.
3	**Iterative**	Everything is moderate: periodic reviews, moderate change with a stable core, 15 developers, and the text says "regular iterations to refine the product". It is larger than what suits agile, and it has change that plan-driven cannot absorb.
4	**Plan-Driven**	The requirements are stable and documented, reviews happen only at milestones, and regulatory compliance needs documentation and a structured process.